

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 3rd Semester End-Term Examination, December 2022
Discrete Structures (CS - 2101)

Time: 3 hours

Full Marks: 50

Write all parts of the same question together.

Marks will be deducted if answers are vague and intermediate steps are skipped

1. (a) Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the (i) reflexive closure of R ; (ii) symmetric closure of R
- (b) Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if (i) a is no shorter than b ? (ii) a weighs more than b ? (iii) $a = b$ or a is a descendant of b ? (iv) a and b do not have a common friend?
- (c) What are the contrapositive and inverse of the conditional statement "I come to class whenever there is going to be a test."?
- (d) Consider the following logical inferences:
 $I1$: If it rains, then the cricket match will not be played. The cricket match was played. There was no rain.
 $I2$: If it rains then the cricket match will not be played. It did not rain. The cricket match was played.
 Which of the following is TRUE (give reasons)?
 (i) Both $I1$ and $I2$ are correct inferences; (ii) $I1$ is correct but $I2$ is not a correct inference;
 (iii) $I1$ is not correct but $I2$ is a correct inference; (iv) Both $I1$ and $I2$ are not correct inferences.
- (e) How many solutions are there to the inequality $x_1 + x_2 + x_3 \leq 11$, where x_1, x_2 , and x_3 are non-negative integers? [hint: introduce an auxiliary variable x_4 such that $x_1 + x_2 + x_3 + x_4 = 11$.]
- (f) A company stores products in a warehouse. Storage bins in this warehouse are specified by their aisle, location in the aisle, and shelf. There are 50 aisles, 85 horizontal locations in each aisle, and five shelves throughout the warehouse. What is the least number of products the company can have so that at least two products must be stored in the same bin?
- (g) What is a lattice? Justify if the posets with the following Hasse diagrams are lattices.

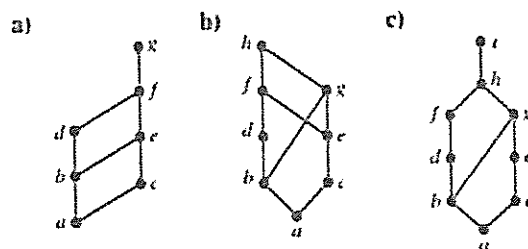


Figure 1: Hasse Diagrams

$$[2 + 4 + 2 + 2 + 3 + 3 + 4 = 20]$$

2. Answer any *two* of the following.

- (a) Prove by mathematical induction that a set with n elements has $\frac{n(n-1)(n-2)}{6}$ subsets containing exactly three elements whenever n is an integer greater than or equal to 3. [4]
- (b) Prove by using strong induction that a simple polygon with n sides, where n is an integer with $n \geq 3$, can be triangulated into $n - 2$ triangles. [4]

- (c) Use mathematical induction to prove that 21 divides $4^{n+1} + 5^{2n-1}$ whenever n is a positive integer. [4]

3. Answer any *two* of the following.

- (a) Draw the *Hasse diagram* for the "less than or equal to" relation on $\{0, 2, 5, 10, 11, 15\}$. Find the maximal, minimal, greatest, and least elements from it. [1 + (1 × 4) = 5]
- (b) Use the *zero-one matrix-based algorithm* to find the transitive closures of the relation $\{(1, 2), (2, 1), (2, 3), (3, 4), (4, 1)\}$ on the set $\{1, 2, 3, 4\}$. [5]
- (c) Use *Warshall's algorithm* to find the transitive closure of the relation $\{(a, c), (b, d), (c, a), (d, b), (e, d)\}$ on the set $\{a, b, c, d, e\}$. [5]

4. Answer any *two* of the following.

- (a) Determine whether the relation R represented by the matrix given below is reflexive, ir-reflexive, symmetric, anti-symmetric, and/or transitive [give brief justifications]. [1 + 1 + 1 + 1 + 2 = 6]

$$R = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

- (b) Determine whether the relations with the directed graph shown below are equivalence relations (give brief justification). [2 + 2 + 2 = 6]

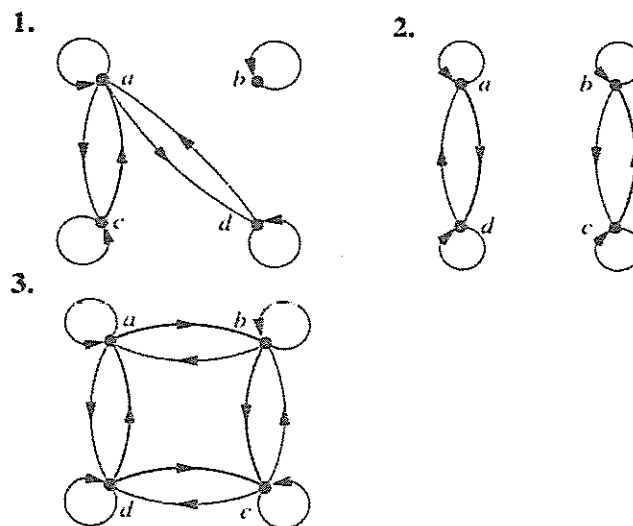


Figure 2: Directed Relation Graphs

- (c) What do you mean by the cardinality of a set? How can we determine if an infinite set is countable or not? Show whether the set $A = \{0, 2, 4, 6, 8, \dots\}$ is countable or not. [1 + 1 + 4 = 6]